25901313 MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER, Discipline Specific Elective (DSE) - MG4DSEHIS202 –

FIELD TECHNIQUES IN ARCHAEOLOGY (2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Part A

Multiple Choice Questions Answer any 10 questions Each question carries 1 marks

1. Thematic map focuses on: Answ: a). One feature
2. Viewshed analysis shows: Answ a). Visible area
3. Vernier caliper measures: Answ a). Thickness
4. Half-life means: Answ a). Time taken for half decay
5. Chronology means: Answ a). Arrangement in time order
6. Probe survey detects: Answ a). Subsurface feature
7. Digital archiving preserves: Answ a). Records
8. Comparative study uses: : Answ a).Similar sites
9. Dendrochronology works best in: Answ a). Temperate regions
10. Amino acid dating is used for: Answ a). Organic remains
11. Scientific sampling increases: Answ a). Reliability
12. Plan photo shows: Answ a). Top view

Part B

Short Answer Questions Answer any 5 questions Each question carries 2 marks

13. What are the main sections of an excavation report?
    Introduction, site location, objectives, methodology, stratigraphy, description of findings, artefacts, illustrations/maps, analysis, interpretation, conclusion, and bibliography. These sections systematically present excavation data, scientific observations, and historical interpretations for proper documentation and future archaeological reference.
14. What is stratigraphic context?
    Stratigraphic context refers to the position of artefacts, features, or deposits within soil layers at an archaeological site. It helps archaeologists determine the chronological sequence of human activities based on the principle that lower layers are generally older than upper layers, ensuring accurate interpretation and dating of archaeological remains.
15. What is sample collection?
    Sample collection in archaeology is the systematic gathering of soil, charcoal, bones, seeds, pottery, or other materials from excavation sites for scientific analysis. It helps

in dating, environmental reconstruction, dietary studies, and understanding past human activities while ensuring accurate recording and preservation of archaeological evidence for future research

16. What are primary sources in archaeological research?

Primary sources in archaeological research are original materials directly related to the past, such as artefacts, inscriptions, monuments, coins, pottery, manuscripts, bones, and excavation records. They provide first-hand evidence about ancient cultures, human activities, chronology, economy, religion, and social life, forming the basic data for archaeological interpretation and analysis.

17. What is LiDAR?

LiDAR (Light Detection and Ranging) is a remote sensing technology that uses laser pulses to measure distances and create detailed three-dimensional maps of the earth's surface. In archaeology, LiDAR helps detect buried structures, ancient settlements, roads, and landscapes hidden beneath vegetation, improving site discovery and archaeological survey accuracy.

18. Why is north arrow used?

A north arrow is used in archaeological drawings to indicate the correct geographical orientation of a site, trench, feature, or artefact. It helps researchers accurately understand spatial relationships, compare maps and plans, and maintain consistency in documentation, interpretation, and future archaeological studies or site reconstruction work.

19. What is a datum point?

A datum point is a fixed reference point used in archaeological excavation for measuring horizontal and vertical positions of trenches, layers, and artefacts. It ensures accurate recording of depth, location, and spatial relationships throughout the excavation process, helping maintain consistency and precision in archaeological mapping, documentation, and interpretation of findings.

20. How does probe survey reduce damage?

A probe survey reduces damage by examining limited portions of an archaeological site before large-scale excavation. It helps identify the presence, depth, and nature of cultural remains with minimal disturbance. This method allows archaeologists to plan excavation carefully, preserve important contexts, and avoid unnecessary destruction of archaeological evidence and structures.

Part C

Short Essay Questions Answer any 4 questions Each question carries 5 marks

21. Explain the methods of surface survey.

Surface survey in archaeology includes systematic walking, field observation, collection of surface artefacts, aerial photography, satellite imagery, and GPS mapping. Archaeologists identify pottery, tools, structures, and soil changes visible

on the ground surface. Transect survey and grid survey methods are commonly used for accurate recording. Surface surveys help locate archaeological sites, understand settlement patterns, and select areas for excavation with minimum disturbance to cultural remains.

22. Explain acoustic reflection principle

The acoustic reflection principle is a geophysical method in which sound waves are transmitted into the ground or water and reflected back from buried objects, layers, or structures. The returning echoes are recorded and analysed to identify subsurface features. In archaeology, this principle is used in sonar and seismic surveys to detect buried settlements, shipwrecks, walls, and geological formations without excavation.

23. Discuss the importance of sieving

Sieving is an important archaeological technique used to separate small artefacts and ecofacts from excavated soil. It helps recover beads, bones, seeds, pottery fragments, and tiny tools that may otherwise be missed during excavation. Sieving improves accuracy in data collection, supports scientific analysis, and provides better understanding of ancient diet, technology, environment, and cultural activities. It also ensures maximum recovery of archaeological evidence from excavation deposits.

24. Discuss materials suitable for TL.

Thermoluminescence (TL) dating is mainly used for materials that were previously heated at high temperatures. Suitable materials include pottery, terracotta, bricks, burnt stones, hearths, kiln remains, and fired clay objects. Certain minerals such as quartz and feldspar are also used. TL dating measures trapped electrons released during reheating and helps determine the time elapsed since the material was last heated. It is useful for dating prehistoric and ancient cultural remains.

25. Describe landscape archaeology.

Landscape archaeology is the study of the relationship between human activities and the natural environment over time. It examines settlements, roads, fields, water systems, monuments, and ecological features within a wider geographical area. Using surveys, GIS, aerial photography, and environmental analysis, archaeologists understand how ancient people modified and interacted with landscapes. This approach helps reconstruct patterns of settlement, economy, communication, agriculture, and cultural development across different historical periods

26. Describe the process of digital map preparation

Digital map preparation involves collecting spatial data through surveys, GPS, satellite imagery, aerial photographs, or excavation records. The data are entered into computer software such as GIS for digitisation, layering, scaling, and analysis. Features like contours, structures, roads, and archaeological sites are accurately plotted. Symbols, legends, coordinates, and north arrows are added for clarity. Digital maps improve accuracy, storage, updating, interpretation, and presentation of archaeological and geographical information for research and planning purposes.

27. Explain the working of magnetometer

A magnetometer is a geophysical instrument used to measure variations in the earth's magnetic field caused by buried archaeological features. Objects such as hearths, walls, pits, ditches, and fired structures create magnetic anomalies. The instrument detects and records these changes while surveying the ground surface. The collected data are converted into maps showing subsurface features. Magnetometer surveys are non-destructive, rapid, and useful for locating hidden archaeological remains without excavation or physical disturbance to the site.

Part D

Essay Questions Answer any 2 questions Each question carries 15 marks

28. Describe the principles of stratigraphy and the use of Harris Matrix in archaeological interpretation

Stratigraphy is the study of soil layers or strata in archaeological excavation to understand the chronological sequence of human activities. It is based on the principle of superposition, which states that lower layers are generally older than upper layers. Other principles include original horizontality, association, and stratigraphic succession. Each layer, feature, or deposit represents a separate archaeological context that helps reconstruct past events and occupation phases.

The Harris Matrix, developed by Edward Cecil Harris, is a diagrammatic system used to record and interpret stratigraphic relationships between archaeological contexts. It arranges layers and features in chronological order, showing which deposits came earlier or later. The Harris Matrix helps archaeologists organise excavation data systematically, identify site formation processes, and reconstruct cultural sequences accurately. It is widely used in modern archaeology for excavation recording, analysis, and interpretation of complex archaeological sites.

29. Discuss applications of K-Ar dating method in archaeology

Potassium–Argon (K–Ar) dating is an absolute dating method used to determine the age of volcanic rocks and ash layers associated with archaeological and prehistoric remains. It is based on the radioactive decay of potassium-40 into argon-40 over long periods of time. This method is especially useful for dating very ancient sites beyond the range of radiocarbon dating.

K–Ar dating is widely applied in prehistoric archaeology, palaeoanthropology, and geological studies. It helps establish the age of early human fossils, stone tools, and volcanic deposits associated with hominid activities. Important prehistoric sites in East Africa, such as Olduvai Gorge, have been dated using this method. It also assists archaeologists in constructing chronological frameworks for human evolution and cultural development. Since volcanic materials are common in many archaeological regions, K–Ar dating provides reliable and scientifically accurate dates for ancient archaeological contexts, particularly those older than several hundred thousand years.

30. Evaluate limitations of relative dating methods.

Relative dating methods determine the chronological sequence of archaeological materials without providing exact calendar dates. One major limitation is that they

only establish whether an object or layer is older or younger than another. Methods such as stratigraphy and typology depend heavily on undisturbed contexts; natural disturbances, erosion, animal activity, or human interference may alter original layers and produce inaccurate interpretations.

Relative dating is also influenced by regional cultural variations, making comparison between different sites difficult. Typological dating may become unreliable when artefact styles continue for long periods or are reused by later cultures. These methods cannot independently determine the absolute age of archaeological remains and often require support from scientific dating techniques such as radiocarbon or thermoluminescence dating. In addition, incomplete excavation records and lack of associated materials may reduce reliability. Therefore, although relative dating is fundamental in archaeology, its limitations make it necessary to combine it with absolute dating methods for accurate chronological interpretation.

31. Explain the structure and components of a standard excavation report.

A standard excavation report is a systematic record of archaeological excavation and findings. It generally begins with an introduction explaining the site location, objectives, historical background, and significance of the excavation. This is followed by methodology, describing survey methods, excavation techniques, sampling procedures, recording systems, and scientific analyses used during the investigation. The main body contains detailed descriptions of stratigraphy, trenches, layers, structures, artefacts, ecofacts, and features discovered during excavation. Plans, maps, photographs, section drawings, and tables are included for accurate documentation. Interpretation and analysis explain cultural sequences, chronology, settlement patterns, economy, religion, and other aspect of past human activity.

The report concludes with findings, conclusions, and recommendations for future research or conservation. Bibliography, appendices, catalogues, and excavation registers are also important components. A properly prepared excavation report ensures scientific accuracy, preservation of archaeological data, and future reference for researchers and heritage management.